Ian Mersich, Ph.D.

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RESEARCH PROFILE

Translational biologist integrating computational discovery with molecular and cellular experimentation to study how proteostasis, RNA regulation, and adaptive cell states shape disease and therapeutic response. My work combines exploratory multi-omic analysis with direct experimental validation, with particular strengths in HSP90 isoform and co-chaperone biology, drug resistance, pediatric neuro-oncology, and neurological disease models. I use selective chemical and genetic perturbations to define mechanisms and test whether disease-associated states can be reprogrammed therapeutically. More than 13 years of experience across academic biomedical research and pharmaceutical development.

EDUCATION

Ph.D., Cancer and Cell Biology — University of Cincinnati | Cincinnati, OH **2023**

Graduate Certificate, Integrative Genomics — University of Saint Joseph **2017**

B.S., Biology — Indiana University Bloomington | Bloomington, IN **2012**

ACADEMIC RESEARCH EXPERIENCE

Postdoctoral Research Associate — University of Notre Dame, Warren Center for Drug Discovery / Blagg Lab | Notre Dame, IN **May 2024–Present**

- Lead computational and translational studies of HSP90β-selective inhibitor resistance, HSP90/AHA1 disruption, and broader drug-response programs by integrating chemical perturbation with transcriptomics, metabolomics, pharmacology, and functional genomics.
- Co-led biological and computational evaluation of HSP90/AHA1 disruptors in activated microglia, combining RNA sequencing, inflammatory and redox assays, and projection of perturbation-derived signatures into disease-associated human microglial states.
- Develop reproducible R/Python workflows for multi-omic integration, model-system selection, pathway and network analysis, perturbation signatures, single-cell state analysis, and publication-quality source-data generation.
- Design and execute mammalian cell culture, cloning, biochemical and cellular assays, dose-response studies, LC/MS and metabolomics experiments, and target-engagement studies in collaboration with medicinal chemists.
- Supervise undergraduate and graduate researchers in project planning, experimental design, reproducible analysis, interpretation, figure preparation, and scientific communication.

Research Fellow / Graduate Student Researcher — Cincinnati Children's Hospital Medical Center / University of Cincinnati | Cincinnati, OH **Jan 2019–May 2024**

Thesis advisor: Biplab Dasgupta, Ph.D., M.B.A.

- Led an integrated epigenetic-metabolism program in H3K27M-mutant diffuse midline glioma using patient-derived models, transcriptomic and chromatin analyses, metabolomics, stable-isotope tracing, pharmacology, and orthotopic mouse models.
- Linked H3K27M-driven chromatin remodeling and Nuclear Factor I activity to dysregulated purine metabolism and dependence on de novo purine biosynthesis. Genetic and pharmacologic ATIC inhibition reduced tumor-cell fitness and improved survival in vivo.
- Characterized folate biology and antifolate resistance in diffuse midline glioma and evaluated strategies to resensitize tumor models to therapy.
- Performed molecular and cellular assays, imaging, immunostaining, mouse studies, and PK/toxicology support while developing computational analyses for isotope tracing, drug response, disease-state prioritization, and model selection.

PUBLICATIONS AND ACTIVE MANUSCRIPTS

Accepted Manuscripts

1. Mersich I, Congrove S, Horchar M, Caceres R, Muraleedharan R, Desai J, Desai P, Sallans L, Reisz JA, Grier A, Hass MR, Donmez O, Yin C, Beaber S, Weirauch MT, Kottyan L, Stevenson CB, Sun C, de Blank P, Pillay-Smiley N, Hummel TR, Elumalai N, Tavassoli A, Firestein R, Phoenix TN, D'Alessandro A, Dasgupta B. Nuclear Factor I Connects H3K27M to a Targetable Purine Dependency in Diffuse Intrinsic Pontine Glioma. *Nature Communications*. Accepted August 18, 2026; final volume and article number pending. Earlier preprint: bioRxiv. 2025. doi:10.1101/2025.05.21.655327.
Integrated metabolomic, transcriptomic, chromatin, isotope-tracing, genetic, and orthotopic studies linked H3K27M and Nuclear Factor I activity to dysregulated purine metabolism and dependence on de novo purine biosynthesis. Genetic or pharmacologic inhibition reduced tumor-cell fitness, slowed orthotopic progression, and extended survival in mice.

Published Articles

2. Serwetnyk MA, Tanguturi P, D'Amico T, Morishima Y, Lau M, **Mersich I**, Barlow D, Houseknecht KL, Osawa Y, Streicher JM, Ali A, Blagg BSJ. Chimeric Hsp90β-Selective Inhibitors Confer Enhanced Affinity, Selectivity, and Drug-Like Properties. *Journal of Medicinal Chemistry*. 2026. Published online October 8, 2026. doi:10.1021/acs.jmedchem.6c02324.

3. Reynolds TS, **Mersich I**, Blagg BSJ. Cell-Impermeable Hsp90α-Selective Inhibitor NDNA4 Inhibits Cancer Cell Migration and Targets Extracellular Hsp90α's Critical F-5 Fragment. *ACS Chemical Biology*. 2026;21(4):876-885. doi:10.1021/acschembio.6c00125.

4. Mersich I, Blagg BSJ, Ali A. Multi-omic integration identifies broad drug resistance mechanisms and strategies to therapeutically reprogram cancer cells. *iScience*. 2026;29(1):114293. doi:10.1016/j.isci.2025.114293.
Integrated PRISM drug-response data with transcriptomic, mutation, and metabolomic profiles to define recurrent broad-resistance programs. Perturbagen analysis led to experimental validation that rosiglitazone reduced NFE2L2-associated expression and resensitized resistant cells to chemotherapy.

5. Mersich I, Malmberg A, Anik E, Hassan MS, von Holzen U, Blagg BSJ, Ali A. Synergistic Disruption of Survival and Metastatic Potential in Esophageal Adenocarcinoma Cells Through Combined Inhibition of HIF1α and CD73. *Cancers*. 2025;17(24):4016. doi:10.3390/cancers17244016.

6. D'Amico T, Serwetnyk MA, Dou X, **Mersich I**, Barlow D, Houseknecht KL, Streicher JM, Ali A, Blagg BSJ. Hsp90β-Selective Inhibitors: Probing the Solvent-Accessible Frontier. *ChemMedChem*. 2025;20(24):e202500657. doi:10.1002/cmdc.202500657.

7. Serwetnyk MA, Strunden T, **Mersich I**, Barlow D, D'Amico T, Mishra SJ, Houseknecht KL, Streicher JM, Blagg BSJ. Optimization of an Hsp90β-Selective Inhibitor via Exploration of the Hsp90 N-Terminal ATP-Binding Pocket. *European Journal of Medicinal Chemistry*. 2025;297:117925. doi:10.1016/j.ejmech.2025.117925.

8. Mersich I, Anik E, Ali A, Blagg BSJ. Integrative Multi-Omics Analyses Reveal Mechanisms of Resistance to Hsp90β-Selective Inhibition. *Cancers*. 2025;17(21):3488. doi:10.3390/cancers17213488.
Combined multi-omic analysis with acquired-resistance models to identify cytoskeletal, mitochondrial, redox, kynurenine-AHR, and compensatory Hsp90α programs. Drug-network analysis nominated carboplatin, and combination testing demonstrated synergy in resistant models.

Manuscripts in Revision

9. Mersich I†, Catalfano KC†, Archdeacon J, Lei W, Blagg BSJ. Disruption of the Hsp90/Aha1 Axis Reveals Disease-Relevant Microglial Inflammatory and Redox-Remodeling Programs. In revision following peer review at *GLIA*. †Co-first authors.
Hsp90/Aha1 disruption selectively remodeled inflammatory, Nos2/iNOS-linked, adaptive-stress, and redox programs in activated microglia rather than globally suppressing activation. Projection into a human cross-dementia single-nucleus atlas provided disease-state context for the perturbation signatures.

10. DeCarlo A, Fernandes da Silva C, Gatesman T, **Mersich I**, Calabretta S, Povilaikaite J, MacLeod G, Shen LQ, Harvey M, Yam PT, Aragao L, Tourigny N, Sbergio SG, Tabori U, Dirks P, Hawkins C, Remke M, Reisz JA, Charron F, D'Alessandro A, Angers S, Agnihotri S, Ramaswamy V. Lineage encoded OXPHOS dependency creates a therapeutic vulnerability in medulloblastoma. Manuscript in revision for *Developmental Cell*.

Public Preprint

11. Datta M, Asaadi Tehrani G, Zarodniuk M, Kubick R, **Mersich I**, Gutierrez A, D'Amico T, Sallaberry H, Smith B, Ali A, Blagg B. Dual Epigenetic and Chaperone Inhibition Disrupts Hypoxia Signaling and Tumor Progression in 3-D Models of Triple-Negative Breast Cancer. Research Square. 2026. doi:10.21203/rs.3.rs-10319464/v1. Collaborator laboratory lists the manuscript as under review.

Manuscripts in Preparation

12. Mersich I. Context-dependent transcriptomic signatures for mature microRNA prediction. Manuscript in preparation.
Developing and benchmarking models that infer mature microRNA abundance from bulk RNA-seq using matched mRNA and small-RNA profiles across cancers, including tests of whether biological context improves miRNA-specific feature selection and prediction.

13. Mersich I, Blagg BSJ. The Hsp90 Chaperone Network in Central Nervous System Development: Isoforms, Co-Chaperones, and Developmental Transitions. Manuscript in preparation.

14. Maurya N, Kondam C, **Mersich I**, Barlow D, Houseknecht KL, Ali A, Blagg BSJ. Exploring the Structural Diversification at the Solvent-Exposed Region of the Hsp90β ATP-Binding Pocket: Effects on Binding Affinity, Antiproliferative Activity, and Solubility. Manuscript in preparation.

15. Mersich I, Congrove S, Horchar M, Caceres R, Beaber S, Dasgupta B. Mechanisms of Resistance to Antifolates in DIPG. Manuscript in preparation.

RESEARCH SUPERVISION AND MENTORING

- Currently supervise undergraduate and graduate researchers across experimental and computational projects, including study design, reproducible analysis, biological interpretation, figure development, and scientific communication.
- Supervised undergraduate researchers and technical staff during doctoral training in pediatric neuro-oncology and cancer-metabolism projects.

AWARDS AND HONORS

- Selected oral presenter and travel-grant recipient, 13th Annual Harper Cancer Research Day, 2025.
- Third-place poster award, University of Cincinnati Graduate School Research Forum, 2022.

SELECTED PRESENTATIONS

- DIPG-17. Dysregulated purine metabolism in a lethal childhood brainstem tumor reveals new treatment strategies with old drugs. *Neuro-Oncology*. 2024.
- TMET-04. Sensitizing DIPGs to antifolates: a preclinical study on purine biosynthesis targets. *Neuro-Oncology*. 2023.
- TMET-35. One carbon and nucleotide metabolism are novel targets in pediatric brainstem tumors. *Neuro-Oncology*. 2022.
- EXTH-40. Therapeutically targeting de novo purine biosynthesis in diffuse intrinsic pontine glioma. *Neuro-Oncology*. 2020.

TECHNICAL AND SCIENTIFIC EXPERTISE

- Computational and genomic analysis — R, Python, bulk and single-cell transcriptomics, multi-omic integration, CRISPR/RNAi dependency analysis, DepMap/PRISM, pathway and network analysis, perturbation signatures, disease-atlas integration, biomarker discovery, model-system selection, and reproducible workflows.
- Chemical biology and therapeutic response — Selective genetic and small-molecule perturbation, dose-response and combination pharmacology, target engagement, drug resistance, HSP90 isoform and co-chaperone biology, mechanism-of-action studies, and medicinal-chemistry collaboration.
- Molecular and cellular biology — Mammalian cell culture, cloning, qPCR, western blotting, biochemical and cellular assays, reporter assays, imaging, immunostaining, experimental design, and mechanistic rescue strategies.
- Analytical and preclinical methods — LC/MS and metabolomics, HPLC, stable-isotope tracing, in vivo cancer models, mouse handling, bioluminescent imaging, PK/toxicology support, analytical method development, qualification, and validation.

PRIOR PHARMACEUTICAL DEVELOPMENT AND INDUSTRY EXPERIENCE

Development Associate — CustoPharm, Inc. | Carlsbad, CA **Jan 2017–May 2018**

- Developed, optimized, transferred, qualified, and validated analytical methods under GMP and authored protocols and reports supporting formulation, stability, and regulatory-filing activities.

Research Associate, Analytical Operations — Gilead Sciences | Oceanside, CA **Oct 2015–Jan 2017**

- Supported process characterization and validation of therapeutic monoclonal antibodies using HPLC, capillary electrophoresis, gel-chip, and ELISA assays while coordinating laboratory operations and instrument readiness.

Scientist I / QC Chemist — Boehringer Ingelheim | Columbus, OH **Feb 2013–Jul 2015**

- Performed analytical release testing, maintained controlled worksheets and LIMS records, peer-reviewed analytical documentation, and supported audit and quality-system requirements.

Laboratory Assistant — Culture Systems, Inc. | Mishawaka, IN **Aug 2012–Feb 2013**

- Performed wet chemistry, reagent preparation, cell counting, and quality-control testing of probiotic cultures and finished products.